

PROF909 — Professional Capstone Project
Assessment 3 — Project Proposal

Recovering Missed-Call Leads for Australian Sole-Trader Trades Businesses

An AI-Assisted SMS Text-Back and Follow-Up Agent

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Pathway: Startup MVP

Sponsor / mentor: To be confirmed (industry mentor sought — see §11, §14)

Public repository: github.com/<your-username>/missed-call-recovery-agent

Word count: ≈ 3,700 (body §2–§15; excludes tables, figures, references and appendices)

Submission date: Part A, Week 10

AI Use Statement: *AI tools assisted with structuring, critique and reference formatting only; all analysis, framework insight and gap reasoning are the author's own. Full statement at Appendix E.*

2 Executive summary

Australia has roughly 1.7 million sole traders, and 97.3% of actively trading businesses are small (ABS 2025; ASBFEO 2025). Many are tradespeople who cannot answer the phone while they are on the tools. For them a missed call is not a delayed conversation; it is a lost job that goes to the next competitor, because most callers who reach voicemail never ring back.

This project will design, build and evaluate a compliance-first SMS text-back and follow-up agent for Australian sole-trader trades businesses. When a genuine inbound call is missed, the agent replies within sixty seconds with a compliant message that identifies the business and offers a one-tap opt-out, runs a short follow-up, and captures the enquiry as a booking — all set up from a phone in under five minutes.

The evidence is settled. Speed of response drives conversion (Oldroyd, McElheran & Elkington 2011); narrowly scoped automation improves service when it escalates gracefully to a human (Ranieri et al. 2024); and for micro-firms the barrier to adoption is effort and trust, not cost (Chouki et al. 2020). Existing tools such as Podium and HighLevel are US-generic, priced for multi-staff firms, and carry no native Australian Spam Act or Privacy compliance — leaving the compliance-first, sole-trader-priced position empty.

Part B will deliver a working MVP and a four-week pilot with five tradies, evaluated against defined success metrics, within a solo budget of approximately 300 hours. The ask is sponsor sign-off, a compliance mentor to review the messaging before go-live, and access to five pilot participants.

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4 Introduction

Australia's smallest businesses are the base of its economy. In June 2025, 97.3% of actively trading businesses were small and about 1.7 million were sole traders operating with no employees (ABS 2025; ASBFEO 2025). A large share are tradespeople — electricians, plumbers, carpenters — who spend the working day physically unable to reach a ringing phone. When the call goes to voicemail, the caller rarely leaves a message; they dial the next business in the search results, and the job is gone.

This matters more in 2026 than before. Cost-of-living pressure has tightened discretionary household spending on trades, so every genuine enquiry carries more weight. At the same time, low-code automation and inexpensive large-language-model inference have put capabilities once reserved for resourced firms within reach of a one-person business. The open question is no longer whether a tool can text a missed caller — that is commoditised — but whether a sole trader will trust it, adopt it in minutes, and operate it safely inside Australia's spam and privacy law.

This proposal sets out a project to close that gap: a compliance-first SMS text-back and follow-up agent scoped specifically for Australian sole-trader trades. It builds on the Assessment 1 evidence review, refines the gap analysis in response to marker and panel feedback, and expands the source base to strengthen the methodology, ethics and sustainability arguments.

The document is written to be read by a sponsor. It distils the evidence and names the gap (§5); states the problem, aim and objectives (§6–§7); specifies the solution and its delivery method (§8–§9); and then commits to a plan, a risk position, an ethical stance, a sustainability case and a stakeholder plan (§10–§14) before bridging to Part B (§15).

5 Evidence review summary and gap analysis

The Assessment 1 review synthesised evidence across five themes. This section distils it and sharpens the claims in response to marker feedback, which noted that some quantified effects rested on practitioner estimates rather than primary data; those figures are now labelled directional and the argument is anchored on peer-reviewed and official sources.

Speed of response

The strongest primary evidence remains Oldroyd, McElheran and Elkington's (2011) audit of 2,241 firms: contacting a web lead within an hour made a qualifying conversation about seven times more likely than waiting longer, and sixty times more likely than waiting a day. Vendor “five-minute rule” multipliers point the same way but are directional and treated as such. That literature assumes a resourced sales team; the sole-trader case, where the binding constraint is physical availability rather than response strategy, is under-theorised — a gap this project addresses empirically through its pilot.

Automated messaging

Whether automation helps or harms service depends on design. A review of forty studies links efficiency and availability to satisfaction but poor intent-understanding to lost trust (Nicolescu & Tudorache 2022); an analysis of 7,167 conversations finds automation reduces effort when it grasps intent and increases it when it does not (Ranieri et al. 2024). A micro-enterprise case study shows throughput gains only where scope was narrow and human escalation was available (Marcineková et al. 2025). The design implication is decisive: scope discipline is the success condition, not a limitation.

Adoption, trust and compliance

For the smallest firms the binding barrier is not cost but effort expectancy — time, confidence and digital literacy (Venkatesh et al. 2003; Chouki et al. 2020; Ayinaddis 2025). In a one-person business the entire learning capacity is

the owner, so five-minute setup and sensible defaults are a necessity, not polish (Cimino et al. 2025). Outbound SMS is governed by the Spam Act 2003 (Cth), which mandates consent, sender identification and a working unsubscribe with substantial penalties for breach (ACMA 2024); caller data is governed by the Australian Privacy Principles (OAIC 2024). A missed inbound call is a form of inferred consent, but the sender must be able to demonstrate it, and giving users control restores trust after error (Dietvorst, Simmons & Massey 2015, 2018). Compliance is therefore a trust-preserving design feature and an administrative burden the tool should carry automatically.

Market and the triangulated gap

Missed-call text-back is commoditised (Podium 2024; HighLevel 2024), so differentiation cannot rest on the feature itself; incumbents are packaged for multi-staff firms, are US-generic, and carry no native Spam Act/APP compliance or trades job-intake. Three gap-analysis frameworks were applied to locate the opportunity. Jobs-to-be-Done showed the job is emotional and reputational (“don’t make me look unprofessional or spammy”), not merely functional. The Value Proposition Canvas showed fit is created by relievers — effort and trust — not features. The Competitor Feature Matrix showed the resulting position — compliance-first, sole-trader-priced, five-minute setup, Australian-trades-aware — is empty in the current market. Three lenses converge on one under-served operator, and that convergence is the design brief for Part B.

6 Problem statement

The problem is concrete and bounded. Australian sole-trader tradespeople lose paid work every week because they cannot answer inbound calls while working, and the callers they miss do not leave voicemail — they call the next business. The loss is not marginal: trades jobs are rarely repeat, often worth several hundred dollars each, and seldom recovered once the caller moves on.

Who has it. The roughly 1.7 million sole traders operating without employees (ABS 2025), concentrated in trades where the operator is physically on the tools for most of the working day. For this group the constraint is availability, not sales process — precisely the case that existing tools and the academic literature do not address.

Scope. The problem this project solves is the recovery of genuine missed inbound enquiries and their conversion into bookings, within Australian spam and privacy law, at a price and setup effort a one-person business can absorb. It explicitly excludes broader CRM, invoicing, scheduling, multi-language and voice-AI needs, which are out of scope for the MVP.

Why now. Cost-of-living pressure raises the value of each recovered enquiry, while cheap automation and LLM inference make instant, compliant response feasible for a solo operator for the first time. The enabling technology exists; what is missing is a trustworthy, compliance-first tool designed for this specific operator — which is the problem this project resolves.

7 Aim and objectives

Aim. To design, build and evaluate a compliance-first SMS text-back and follow-up agent for Australian sole-trader trades businesses that recovers genuine missed-call enquiries within sixty seconds, sets up in under five minutes, and operates within the Spam Act 2003 (Cth) and the Australian Privacy Principles — demonstrating viability for unassisted adoption by a one-person business.

Objectives (mapped to Part B milestones).

1. **By Part B Week 6**, deliver a working MVP that detects a genuine missed inbound call and sends a compliant text-back — business name, ABN and one-tap opt-out — delivered within 60 seconds in at least 40% of test events (Specific, Measurable, Time-bound). [M1]

2. **By Part B Week 7**, pass a Spam Act / APP compliance review with zero defects against a documented checklist, gating any live messaging (Measurable, Relevant). [M2]
3. **By Part B Week 11**, complete a four-week live pilot with five sole-trader tradies in which at least 15% of texted callers respond or book, with median setup under five minutes across the five users (Specific, Measurable, Achievable, Time-bound). [M3/M4]
4. **By Part B Week 12**, deliver a final evaluation report and public portfolio handover documenting metrics, a pivot-or-persist decision, and maintenance notes (Relevant, Time-bound). [M5]

Each objective names a deliverable, a measurable target and a date, and is achievable within a solo Part B budget of approximately 300 hours.

8 Proposed solution

The artefact is a lightweight, event-driven SMS agent, not a platform. Its job is narrow by design: recover a genuine missed inbound enquiry and turn it into a booking, while carrying the compliance burden for the operator.

Figure 1. MVP system architecture (event-driven, human-in-the-loop)

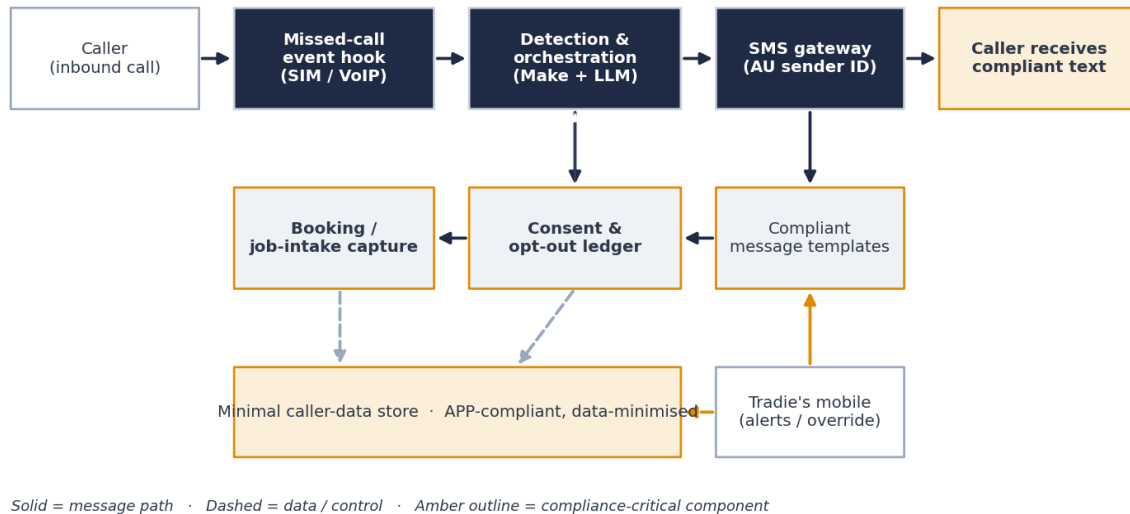
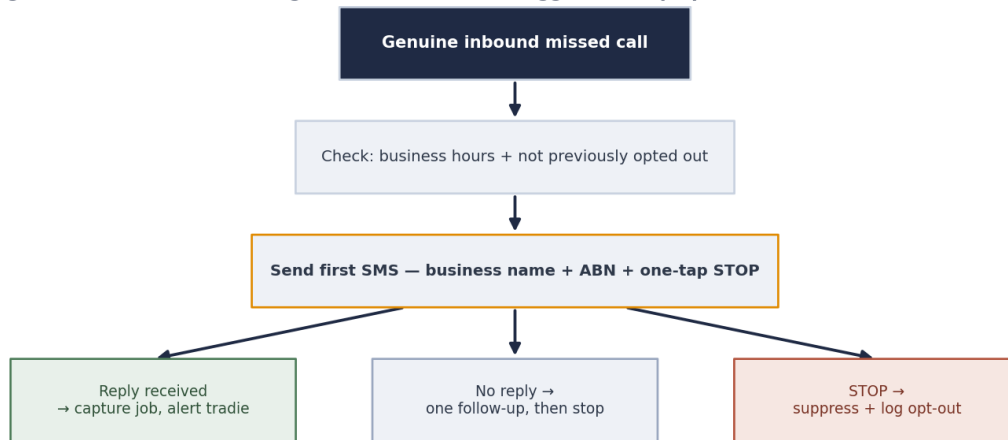


Figure 1. MVP system architecture — the message path runs left to right; compliance-critical components carry an amber outline.

How it works (Figure 1). A missed-call event — captured through a SIM-based or VoIP hook on the tradie's existing number — triggers a detection-and-orchestration layer built on a low-code automation engine (Make) with a large-language-model step for message composition and simple intent handling. Before anything is sent, the orchestrator checks the consent and opt-out ledger and business-hours rules. It then sends a first SMS through an Australian SMS gateway that identifies the business by name and ABN and offers a one-tap STOP. If the caller replies, the agent captures a short structured job intake (what, where, when) and alerts the tradie; if the caller sends STOP, the number is suppressed and the opt-out logged; if there is no reply, a single follow-up is sent and the sequence ends.

Figure 2. Consent-safe message flow (conservative trigger, one-tap opt-out)



Only genuine inbound callers are ever messaged — never a purchased or scraped list.

Figure 2. Consent-safe message flow — only genuine inbound callers are ever messaged, never a purchased or scraped list.

Guardrails are built in, not bolted on. Every message carries identification and opt-out; the tradie has one-tap pause and can take over any conversation at any point; caller data is minimised and stored only as needed under the Australian Privacy Principles (OAIC 2024). These are direct design responses to the trust evidence: control restores confidence after error (Dietvorst et al. 2015, 2018), and conservative triggering keeps the tool inside inferred-consent boundaries (ACMA 2024).

Alignment with the evidence and the gap. Each design choice traces to a finding. Sixty-second response operationalises the speed-to-lead effect (Oldroyd et al. 2011). Narrow scope with graceful human escalation reflects the automation evidence that scope discipline is the success condition (Ranieri et al. 2024; Marcineková et al. 2025). Five-minute phone setup and sensible defaults answer the adoption barrier of effort and confidence (Chouki et al. 2020; Ayinaddis 2025). Native identification, opt-out and data minimisation answer the compliance-and-trust theme (ACMA 2024; OAIC 2024). Together these occupy the empty position the gap analysis exposed: compliance-first, sole-trader-priced, five-minute setup, Australian-trades-aware.

MVP scope is deliberately bounded. In scope: missed-call detection and instant text-back; a short follow-up sequence; consent-safe templates; one-tap pause/override; five-minute setup; English only. Out of scope: full CRM, invoicing and payments, calendar-integrated scheduling, multi-language, voice AI, and multi-staff routing. This boundary is a scope-discipline decision, not an omission — it keeps the build inside the 250–350-hour solo envelope and concentrates effort on the one capability the evidence says will succeed. The substantive Part A artefact accompanying this proposal is the architecture and message-flow design, the consent state model and a minimal data schema, published in the project repository as the foundation for the Part B build.

9 Methodology and delivery approach

This project follows the Lean Startup build–measure–learn cycle (Ries 2011), integrated with design thinking for the discovery phase (Brown 2009) and a light Scrum cadence for delivery rhythm (Schwaber & Sutherland 2020). Figure 3 shows the approach.

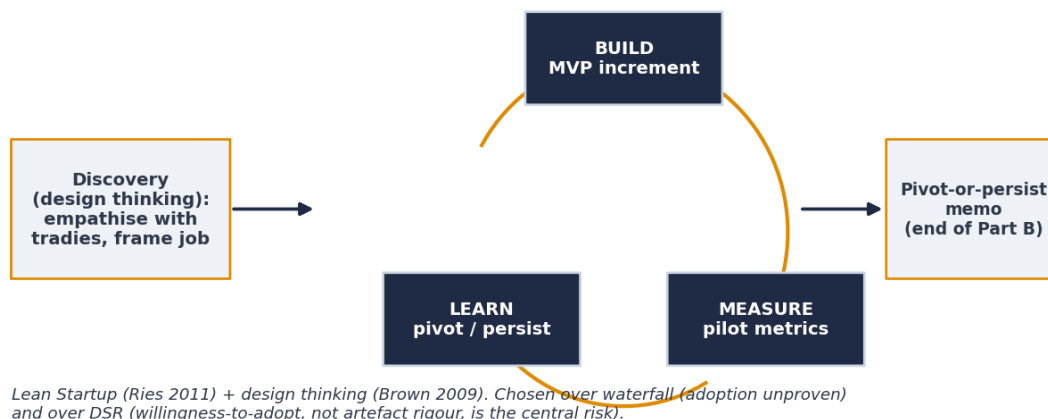
Figure 3. Delivery approach — design-thinking discovery feeding a Build-Measure-Learn loop

Figure 3. Delivery approach — design-thinking discovery feeds an iterative Build-Measure-Learn loop, closing with a pivot-or-persist decision.

Why Lean. The project's central risk is not technical — texting a missed caller is commoditised — but behavioural: whether a sole trader will trust and adopt the tool, and whether recovered enquiries convert. These are unverified hypotheses about willingness to adopt and to book. Lean is preferred over waterfall because building a full product before testing adoption would lock in unvalidated assumptions about the very thing most in doubt; the build-measure-learn loop tests those assumptions with a real pilot at the earliest defensible point. Lean is preferred over Design Science Research (Hevner 2007) because DSR's centre of gravity is rigorous evaluation of an artefact against design objectives, whereas here the decisive question is market-facing — does the operator adopt and does the caller convert — not whether the artefact meets an internal specification. DSR's rigour cycle nonetheless informs the build: the evidence base and compliance checklist act as the knowledge grounding for design decisions.

How it runs. Discovery uses design thinking to empathise with tradies and frame the job to be done, converting the Assessment 1 Jobs-to-be-Done and Value Proposition Canvas insight into concrete message and intake templates. The build then proceeds in short increments, each ending in a measurable check rather than a feature list. Adapted for solo delivery, a light Scrum cadence provides structure: two-week increments with a written self-retrospective and a fortnightly review with the industry mentor stand in for team ceremonies. The measure phase is the four-week pilot; the learn phase is a documented pivot-or-persist decision at the end of Part B.

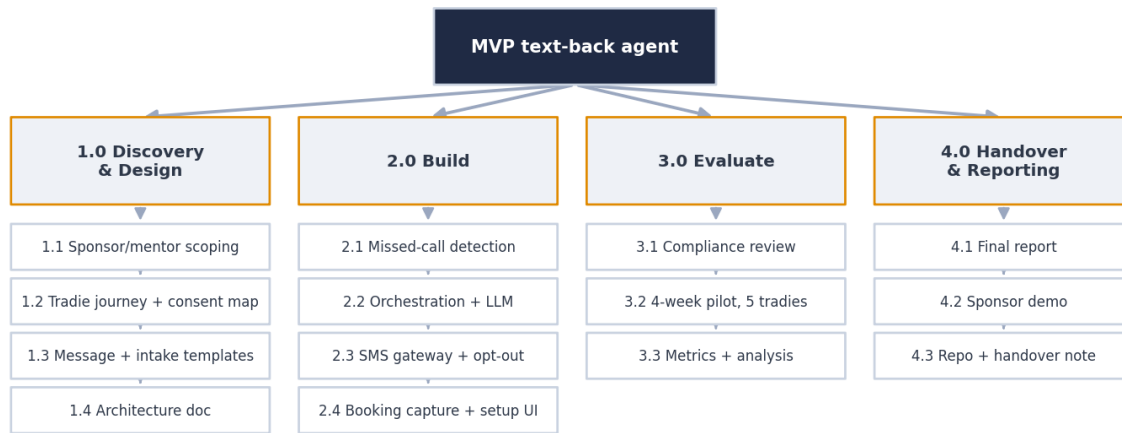
Evaluation design. The pilot is evaluated with a simple mixed approach consistent with Creswell and Creswell (2023): quantitative service metrics (proportion of missed calls texted back within sixty seconds; proportion of texted callers who respond or book; median setup time) alongside short qualitative feedback from the five participating tradies on trust, effort and reputation. Quantitative metrics establish whether the mechanism works; qualitative feedback explains adoption behaviour the numbers alone cannot — matching the evidence that adoption turns on effort and trust rather than features. The Startup MVP pathway commits to shipping and testing a minimum viable artefact, not to a research study or a fully specified enterprise system, so methodology is chosen to match the project's dominant uncertainty rather than assumed by default.

10 Project plan

The plan decomposes into four phases (Figure 4): Discovery and Design, Build, Evaluate, and Handover and Reporting. Each work-breakdown leaf is sized at roughly 10–40 hours; total Part B effort is budgeted at

approximately 300 hours, inside the 250–350-hour solo envelope. Right-sizing is deliberate — the MVP scope boundary in §8 is what makes the plan deliverable by one person in one term.

Figure 4. Work breakdown structure (three levels)



Each leaf ≈ 10–40 h. Total Part B effort budgeted at ~300 hours (within the 250–350 h solo envelope).

Figure 4. Work breakdown structure — three levels, each leaf sized to 10–40 hours.

Work breakdown. Discovery (1.0) confirms scope with the sponsor and mentor, maps the tradie journey and consent flow, drafts message and intake templates, and produces the architecture document. Build (2.0) delivers four technical leaves: missed-call detection; orchestration with the LLM step; the SMS gateway with opt-out handling; and booking capture with the five-minute setup interface. Evaluate (3.0) runs the compliance review, the four-week pilot with five tradies, and metric analysis. Handover (4.0) produces the final report, sponsor demo and repository handover note.

Figure 5. Part B Gantt summary — critical path in amber, buffer in Week 8

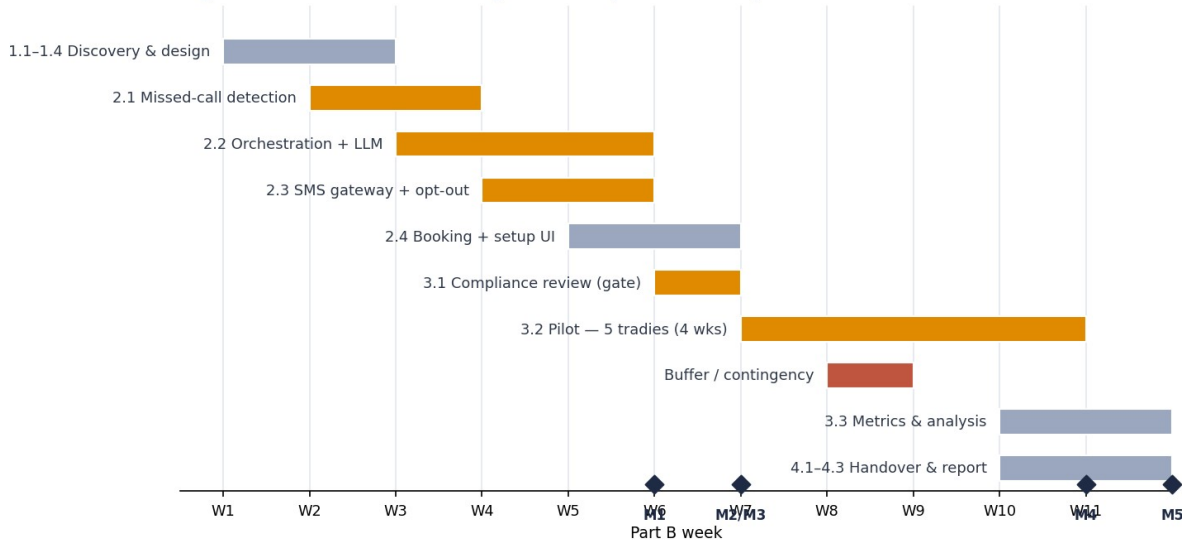


Figure 5. Part B Gantt summary — critical path in amber; buffer week at Week 8 inside the pilot window.

Schedule and critical path (Figure 5). The critical path runs Build → Compliance review → Pilot: no message is sent live until the Spam Act/APP review passes, so the compliance gate sits on the critical path by design. Detection and orchestration (Weeks 2–5) precede the compliance gate (Week 6); the four-week pilot runs Weeks 7–11; analysis

and handover close Weeks 10–12. Milestones are M1 working text-back (W6), M2 compliance sign-off (W7), M3 pilot live (W7), M4 pilot evaluation (W11) and M5 handover (W12).

Contingency and resources. A buffer week at Week 8, inside the pilot window, absorbs recruitment slippage, gateway issues or personal disruption without moving the end date — a direct response to A2 panel feedback that the original timeline carried no slack and compressed compliance into a single week; compliance is now planned as preparation before the gate rather than a single-week task. The build uses low-code tooling and LLM inference, an Australian SMS gateway, and a modest message/inference budget (estimated AUD 150–250 during the pilot). Human resources are the student's ~300 hours, fortnightly industry-mentor review, sponsor scoping and sign-off, and five pilot tradies recruited from Week 5 with a remote-fallback plan.

11 Risk management

Risk is scored on likelihood (L) and impact (I), each 1–3, with priority = $L \times I$; higher scores are managed first. The register below lists the nine most material risks with concrete mitigations and named owners; the full register is at Appendix C and the responsibility split at Appendix D. The dominant risks are not technical but compliance, adoption and personal-capacity risks, which is consistent with the evidence base and concentrates mitigation where it matters. Two mitigations follow directly from A2 panel feedback: a Week-8 buffer for schedule risk, and early pilot recruitment with a remote-testing fallback. The three highest-priority items each carry a specific, actionable mitigation and a single accountable owner.

#	Risk	L	I	Score	Mitigation	Owner
R1	Compliance / spam breach (consent, sender ID, opt-out)	M	H	6	Conservative inbound-only trigger; name + ABN + one-tap opt-out in first message; automated opt-out honouring; compliance-checklist review gate before pilot	Student
R3	Low adoption / setup friction	M	H	6	Five-minute setup tested early with real tradies; sensible defaults; in-context help	Student
R8	Personal — illness or burnout in Part B	M	H	6	Week-8 buffer built into plan; weekly workload self-check; Special Consideration pre-familiarised	Student
R2	Pilot-tradie recruitment delays	M	M	4	Recruitment starts Part B Week 5; sponsor + trade network pre-identified; remote onboarding fallback	Student
R4	SMS gateway deliverability / sender-ID issues	M	M	4	Evaluate two AU gateways; register sender ID early; fallback gateway prepared	Student
R5	LLM misreads caller intent, eroding trust	M	M	4	Narrow templates; confidence threshold triggers human handoff; tradie override	Student
R7	Scope creep (CRM, scheduling, 2nd language)	M	M	4	Scope locked at A3 charter; change request needs mentor sign-off	Student
R6	Privacy incident (caller data)	L	H	3	Data minimisation by design; APP-compliant storage; no caller data to external services beyond gateway; pre-pilot audit	Student + Sponsor

#	Risk	L	I	Score	Mitigation	Owner
R9	Regulatory change (ACMA / OAIC) during Part B	L	H	3	Track ACMA/OAIC updates; monthly sponsor check-in; pivot to research framing if deployment blocked	Student + Sponsor

Table 1. Risk register (summary). Full register with all ten risks at Appendix C.

12 Ethics, privacy and responsible AI

The project is bound by the ACS Code of Professional Conduct — in particular the Primacy of the Public Interest and Honesty — and applies four named frameworks operationally rather than in passing: Australia's AI Ethics Principles (DISR 2024), the Australian Privacy Principles (OAIC 2024), the Spam Act 2003 (Cth) (ACMA 2024), and the NIST AI Risk Management Framework (NIST 2023), with the ACSC Essential Eight (ACSC 2023) informing baseline security.

The central ethical dilemma is consent. An agent that texts every missed caller risks sending unsolicited commercial messages to people who have not meaningfully consented — a Spam Act breach and, worse for a sole trader whose asset is local reputation, a trust-destroying act. The resolution is conservative by design and maps to the AI Ethics Principles. Transparency and explainability: the first message identifies the business by name and ABN, so the recipient knows they are dealing with an automated response. Fairness: only genuine inbound missed callers are messaged — the best available evidence of inferred consent — never a purchased or scraped list, so the tool cannot become a spam channel. Contestability: a one-tap STOP appears in the first message and is honoured automatically, so opt-out carries no friction. This is the conservative-trigger rule the algorithm-aversion evidence supports (Dietvorst et al. 2015, 2018): giving people control preserves trust.

Privacy by design. Caller data is minimised to what the job intake requires, retained only as long as needed, and not sent to external services beyond the SMS gateway; the LLM step composes messages without persisting caller identifiers externally. Responsible-AI practice follows the NIST AI RMF functions — Govern, Map, Measure, Manage: the compliance checklist and consent ledger operationalise Govern and Manage; the pilot metrics operationalise Measure; and a model/prompt card documenting the LLM step's purpose and limits accompanies the repository. Baseline security follows the Essential Eight (timely patching, multi-factor authentication, least-privilege access to the message store).

Accountability. Because the tool automates outbound contact on behalf of a business, accountability is explicit: the tradie remains the accountable sender, can pause or override at any time, and the system logs consent and opt-out events so consent can be demonstrated to a regulator. Compliance here is a first-order design feature, integrated into the architecture, not a paragraph appended after the build.

13 Sustainability and social impact

Economic. The sector is financially vulnerable, with cash-flow and payment stress a documented risk (ASBFEO 2025); recovering otherwise-lost enquiries strengthens the financial resilience of micro-businesses that sustain local employment — the project's central social contribution. **Environmental (Green IT).** The design uses lightweight, event-based messaging rather than always-on generative calls, so compute is incurred only when a real call is missed, and LLM inference is limited to short message-composition steps, keeping the per-event footprint small (Cimino et al. 2025). **Social equity.** The tool deliberately serves the operators least equipped — by time, confidence and budget — to adopt enterprise software, narrowing the capability gap between sole traders and staffed competitors; the five-minute phone setup and the accessibility of the setup interface (targeting WCAG 2.2 AA) are equity measures, not polish.

14 Stakeholder and communication plan

Five stakeholder groups shape the project. Each has a defined communication mechanism — not simply “email” — and a frequency matched to its influence and interest, so engagement is deliberate rather than reactive. The sponsor and industry mentor drive delivery and receive the most frequent, structured contact; the academic supervisor governs rigour; pilot tradies are engaged intensively but briefly during the pilot window; and the regulator sets the compliance boundary the product is designed around.

Stakeholder	Interest	Influence	Communication mechanism	Frequency
Sponsor / industry mentor	High	High	Fortnightly review call + written sprint note; ad-hoc message for blockers	Fortnightly + ad-hoc
Academic supervisor	Medium	High	Scheduled check-in; written summary against milestones	Fortnightly
Pilot tradies (5)	High	Medium	Written consent + onboarding; short structured feedback session	Onboarding + weekly (pilot)
End callers	High	Low	In-message transparency (business ID) + one-tap opt-out	Every interaction
Regulator (ACMA / OAIC)	Medium	High	Monitor published guidance; compliance documented against it	Monthly scan

Table 2. Stakeholder and communication plan (interest × influence).

15 Conclusion and bridge to Part B

This proposal commits to a bounded, evidence-grounded deliverable: a compliance-first SMS text-back agent for Australian sole-trader tradies, built and pilot-tested within a solo budget of approximately 300 hours and defended by a clear risk and ethics position. The evidence identifies an under-served operator; the gap analysis locates an empty market position; the plan turns both into a testable MVP. Part B Week 1 begins with sponsor scoping confirmation and the discovery leaf (1.1–1.2) — the tradie journey and consent map — feeding directly into the first build increment. The foundation, in repository form, is already in place.

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Appendix A Updated Project Charter

Revised from the Assessment 1 charter in response to A1 marker feedback (evidence base and methodology) and A2 panel feedback (timeline buffer, staged compliance, early recruitment).

Project & pathway

AI-Assisted SMS Text-Back and Follow-Up Agent for Australian Sole-Trader Trades Businesses. Student: Shreekrishna Gyawali (SHEA25124). Pathway: Startup MVP.

Problem statement

Australian sole-trader tradespeople miss inbound calls while on the tools, and most callers who reach voicemail do not call back — they move to the next business. Existing recovery tools are priced for multi-staff firms and lack native Australian spam/privacy compliance, leaving the ~1.7 million-strong sole-trader segment unserved.

Value proposition

Recover missed calls as booked work within seconds via compliant SMS text-back, set up in five minutes on a solo budget — protecting both revenue and local reputation.

Scope

In: missed-call detection and instant text-back; short follow-up sequence; consent-safe templates; one-tap pause/override; five-minute phone setup; English-only MVP. **Out:** full CRM, invoicing/payments, calendar scheduling, multi-language, voice AI, multi-staff routing.

Stakeholders

Sole-trader tradesperson (primary user/buyer); end caller (experience and consent); industry mentor (feasibility/go-to-market); academic supervisor (rigour); regulator ACMA/OAIC (compliance boundary).

Success metrics (SMART)

≥40% of missed calls receive a delivered compliant text-back within 60 seconds in pilot; ≥15% of texted callers respond or book over a four-week pilot; median setup under 5 minutes across 5 test users; zero defects in a Spam Act/APP compliance review.

Top three risks (with mitigations)

- **Compliance breach** — conservative inbound-only trigger, ABN identification and one-tap opt-out, checklist review gate before any live message.
- **Low adoption / setup friction** — five-minute setup tested early with real tradies; sensible defaults.
- **Schedule / capacity** — Week-8 buffer added and compliance staged as preparation before the gate (A2 feedback); pilot recruitment begins Week 5.

Appendix B Detailed Gantt chart

Part B, Weeks 1–11 (handover completes W11–12). Amber cells mark the critical path (Build → Compliance gate → Pilot); the Week-8 buffer sits inside the pilot window. Milestones: M1 text-back working (W6); M2 compliance sign-off (W7); M3 pilot live (W7); M4 evaluation (W11); M5 handover (W12).

Task (WBS)	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11
1.0 Discovery & design											
2.1 Missed-call detection											
2.2 Orchestration + LLM											
2.3 SMS gateway + opt-out											
2.4 Booking + setup UI											
3.1 Compliance review (gate)											
3.2 Pilot — 5 tradies (4 wks)											
Buffer / contingency											
3.3 Metrics & analysis											
4.1–4.3 Handover & report											

Table B1. Detailed Part B schedule by WBS task and week (readable export; also mirrored in the GitHub project board).

Appendix C Risk register (full)

Likelihood (L) and impact (I) scored 1–3 (Low/Medium/High); Score = L × I. Higher score = higher priority.

#	Risk	L	I	Score	Mitigation	Owner
R1	Compliance / spam breach (consent, sender ID, opt-out)	M	H	6	Conservative inbound-only trigger; name + ABN + one-tap opt-out in first message; automated opt-out honouring; compliance-checklist review gate before pilot	Student
R2	Pilot-tradie recruitment delays	M	M	4	Recruitment starts Part B Week 5; sponsor + trade network pre-identified; remote onboarding fallback	Student
R3	Low adoption / setup friction	M	H	6	Five-minute setup tested early with real tradies; sensible defaults; in-context help	Student
R4	SMS gateway deliverability / sender-ID issues	M	M	4	Evaluate two AU gateways; register sender ID early; fallback gateway prepared	Student
R5	LLM misreads caller intent, eroding trust	M	M	4	Narrow templates; confidence threshold triggers human handoff; tradie override at any point	Student
R6	Privacy incident (caller data)	L	H	3	Data minimisation by design; APP-compliant storage; no caller data to external services beyond gateway; pre-pilot audit	Student + Sponsor
R7	Scope creep (CRM, scheduling, 2nd language)	M	M	4	Scope locked at A3 charter; change request needs mentor sign-off	Student
R8	Personal — illness or burnout in Part B	M	H	6	Week-8 buffer built into plan; weekly workload self-check; Special Consideration pre-familiarised	Student
R9	Regulatory change (ACMA / OAIC) during Part B	L	H	3	Track ACMA/OAIC updates; monthly sponsor check-in; pivot to research framing if deployment blocked	Student + Sponsor
R10	Industry mentor unavailable mid-Part B	L	M	2	Secondary mentor identified via ACS Sydney chapter; facilitator can step in for technical review	Student

Table C1. Full risk register (ten risks).

Appendix D RACI matrix

R = Responsible (does the work); A = Accountable (one person, signs off); C = Consulted (input); I = Informed. Each row has exactly one A.

Activity	Student	Industry mentor	Sponsor	Academic sup.	Regulator
Discovery & scoping	R/A	C	C	I	I
Architecture & methodology design	R/A	C	I	C	—
Build MVP	R/A	C	I	I	—
Compliance review (Spam / APP)	R	C	A	I	C
Pilot execution	R/A	C	C	I	I
Privacy & data handling	R	I	A	I	C
Evaluation report	R	C	I	A	—
Sponsor demo & handover	R	C	A	I	—

Table D1. RACI matrix — activities × stakeholders.

Appendix E AI Use Statement

Tools and purposes

- **Claude (Anthropic)** — structuring the proposal against the brief, critiquing the strength of synthesis in draft sections, and checking Harvard reference formatting for consistency. Used for organisation and critique, not for analytical content.
- **Web search (via Claude)** — to locate candidate academic and industry sources; each source was then opened and verified individually against the publisher's record.
- **Grammarly** — proofreading the final draft for grammar and clarity only (no content suggestions accepted).

Representative prompts

- “Here is my Assessment 1 evidence review and the A3 brief's 16-section structure. Where is my gap analysis summarised rather than argued? Do not rewrite it.”
- “Check this Harvard reference list for consistency of style and flag inconsistencies — do not add or change references.”
- “Critique whether my methodology section justifies Lean Startup against alternatives, or merely names it.”

Verification

- Every source was read and its author, year, outlet and claim verified against the publisher record before inclusion.
- All quantified effects were checked; where primary data were unavailable, figures are labelled directional in the text.
- Framework application, gap reasoning, scope decisions and the project plan are the author's own work.

What AI was not used for

AI was not used to perform the gap analysis, write the charter or executive reasoning, or generate the analytical argument of any section. All reasoning, framework insight and identification of the gap are the author's own. This submission reflects the author's judgement — including when to use AI and when not to — rather than AI output.